Each block node has a complete copy of the distributed ledger within a blockchain network. As illustrated in Figure 3, a valid block hash begins with four leading zeros, indicating its validity. Each new hash is generated by combining the previous hash, the new transaction block, and a nonce. The difficulty of the hash, determined by the number of leading zeros it must have, regulates the speed of block production to approximately one block every 10 minutes. A higher difficulty makes it more challenging for anyone to alter the blockchain and use it for double spending.

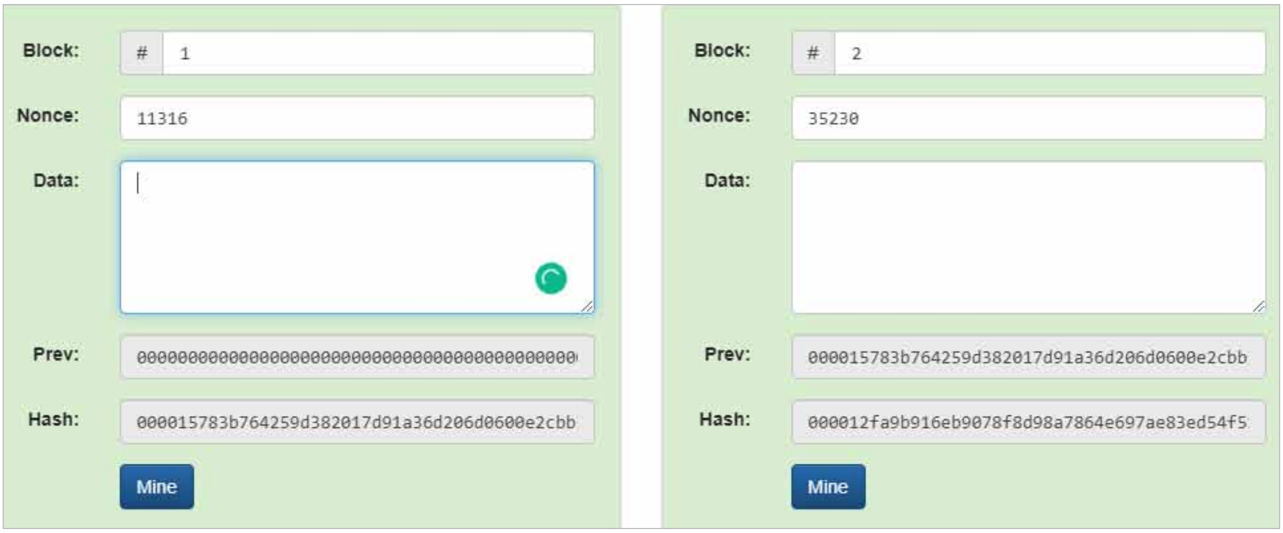

Figure 3: Demonstration of blockchain working on distributed nodes

Table 2: Popular blockchain platforms

Criteria	Ethereum	Hyperledger Fabric	R3 Corda	Quorum
Type	Public	Permissioned	Permissioned	Permissioned
Target industry	General-purpose blockchain platform	Various, but primarily enterprise	Financial services, supply chain, healthcare	Financial services, enterprise
Programming built-in	Solidity	Go, Java, JavaScript	Java, Kotlin	Solidity, Go, Java
Governance	Ethereum developers	Linux Foundation	R3 Consortium	J.P. Morgan and network participants
Cryptocurrency used	Ether	Can integrate with various cryptocurrencies	Not applicable (permissioned network)	Quorum ERC-20 tokens, but can incorporate other cryptocurrencies
Privacy feature	Limited	Supported	Supported	Supported
Consensus mechanism	Proof of Work (PoW)	Pluggable framework	Pluggable (supports various consensus algorithms)	Pluggable (based on Ethereum)
Complexity	High complexity	Moderate-- requires understanding of Fabric architecture	Moderate-- requires understanding of Corda flows and states	Moderate--requires understanding of Ethereum-based architecture
Transaction fees	Gas fees vary based on network demand	Configurable by network operators	Not applicable (permissioned network)	Configurable by network administrators

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